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Report On

**Missing Values Imputations in functional data using
Gaussian Process**

by

M.Sc. II [2024-2025]

DEPARTMENT OF STATISTICS



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CERTIFICATE OF COMPLETION OF RESEARCH PROJECT-II

This is to certify that Mr./Miss.....
of M.Sc. (Statistics) Part-II Sem-IV has satisfactorily completed the
work in Research Project-II during February, 2025 –May, 2025.

Date:

Teacher In-charge

Head,
Department of statistics

Acknowledgment

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Finally, we express our deepest appreciation to all the team members of this review project for their dedication, collaboration, and hard work, which made this project a reality.

Regards,

Miss Tazeen Chanda

Miss Dwarka Jadhav

Miss Sharayu



Declaration

We, The students of Department of Statistics, Punyashlok Ahilyadevi Holkar Solapur University, Solapur, hereby affirms that the project titled "Missing Value Imputation of Functional Data using Gaussian Process", submitted to Punyashlok Ahilyadevi Holkar Solapur University, Solapur, is presented as a requirement for the IV-Semester assessment. This report reflects genuine and original work completed by us under the guidance of **Miss Khushi Khadloya**, Assistant Professor in the Department of Statistics, Solapur University, Solapur. Furthermore, We confirm that this work has not been submitted to any other institution for the award of a degree, diploma, or fellowship.

Regards,

Tazeen Chanda

Dwarka Jadhav

Sharayu Jagtap

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Abstract

This study explores the imputation of missing values in functional data using Gaussian Process (GP), with the inbuilt dataset in R Software. Functional data, characterized by smooth, continuous observations over a domain, poses challenges for traditional imputation techniques due to its infinite-dimensional nature. We employ GP models with a Radial Basis Function (RBF) kernel to exploit temporal correlations and predict missing segments of growth trajectories. The method models the underlying function as a draw from a GP prior and computes the posterior distribution conditioned on observed data. Performance is evaluated using Root Mean Square Error (RMSE) and graphically by comparing the graphs of original v/s the imputed curves, showing that the GP-based approach accurately recovers the missing values while preserving smoothness and structure. Our results demonstrate the robustness and flexibility of Gaussian Processes in functional data analysis, offering a probabilistic framework with well-quantified uncertainty for imputation tasks.

Introduction

Functional data refers to data that are collected over a continuous domain such as time, space, or frequency. Instead of recording individual values at separate points, we observe entire curves, functions, or trajectories. Each of these curves is treated as a single data unit.

Functional Data Analysis (FDA) is a statistical method designed to handle such data. It focuses on analyzing smooth functions instead of raw, discrete measurements. The idea is that real-world processes often change continuously, and treating them as smooth functions gives a more accurate and complete picture.

Importance of Functional Data Analysis

Functional Data Analysis (FDA) is highly valuable when analyzing patterns that evolve continuously over time, space, or frequency. Unlike traditional methods that focus on individual data points, FDA allows us to study entire curves or trajectories, providing a clearer and more detailed understanding of how a process changes smoothly.

Functional data plays a crucial role in modern data analysis, especially in fields where continuous monitoring is common, such as healthcare, finance, environmental science, and traffic systems. Its main strength lies in its ability to capture the complete shape and

behavior of a phenomenon over a domain, offering a richer and more informative representation than discrete point-based data.

Why Choose Functional Data over Time Series and Multivariate Data?

In statistical analysis, choosing the appropriate data structure is crucial for accurate modeling and interpretation. Among the commonly used data types are time series data, multivariate data, and functional data. Each of these has its own structure and focus. However, when the goal is to study processes that vary smoothly and continuously over a domain such as time or space, Functional Data Analysis (FDA) becomes a more suitable and effective approach.

How Functional data looks?

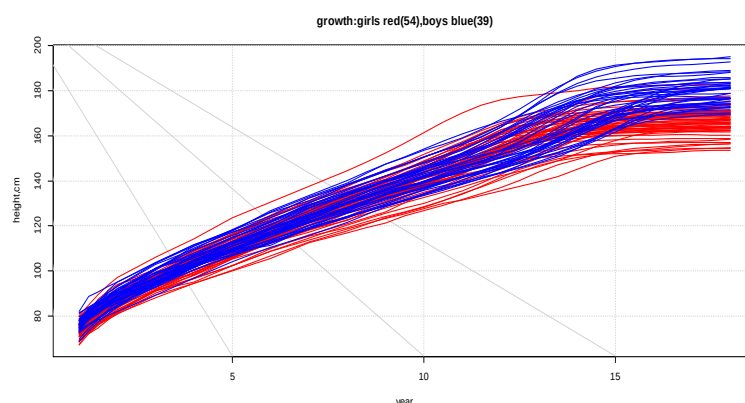


Figure 1:Functional data(Growth data)

Missing Values in Functional Data

Statistical analysis generally provides reliable and meaningful results when the data is complete. However, in real-world scenarios, missing data is a common issue. In the context of functional data, missingness can occur due to various reasons such as human errors, machine malfunction, or difficulties during the data collection process. These gaps can significantly affect the quality of the analysis if not properly addressed.

Missing values in functional data are especially challenging because each observation is a continuous function over a domain like time or space. When parts of a curve are missing, it can distort the overall shape and lead to incorrect inferences. Incomplete curves can also interfere with processes like curve alignment, smoothing, or derivative calculation.

Objectives

- To propose a model for imputing missing values of functional data.
- To enhance the model for more sparse and complex data.
- Predicting the accuracy of our imputations visually by graph.
- Reducing the value of Root Mean Square Error (RMSE)
- Giving the new imputation method which will work better than the single imputation technique which are most common in use.

Types of Imputations

Single Imputation

Single imputation is a statistical technique used to address missing data by replacing each missing value with a single estimated value. This method allows for the transformation of an incomplete dataset into a complete one, which can then be analyzed using conventional statistical procedures that require no missing entries.

Common approaches to single imputation include:

Mean, Median, or Mode Imputation: Replacing missing values with the mean, median, or mode of the observed data for that variable.

Mean and Median Imputation.

Mean and median imputation are basic yet widely used techniques for handling missing data, particularly in numerical variables. These methods fall under the category of single imputation, where each missing value is replaced with a single, computed statistic based on the observed (non-missing) data. They are primarily employed due to their simplicity and ease of implementation.

1. Mean Imputation

Mean imputation involves replacing missing values with the arithmetic mean of the available (non-missing) data for a given variable. This approach assumes that the data are missing completely at random (MCAR) and that the mean is a representative measure of central tendency for the variable.

Let $x_1, x_2, \dots, x_n$ be a numerical variable with some missing values. The mean of the observed values is computed as:

$$\bar{x} = \frac{1}{n_{\text{obs}}} \sum_{i=1}^{n_{\text{obs}}} x_i \quad \dots\dots\dots (1)$$

where n_{obs} is the number of non-missing values in the variable. Each missing value X in is then replaced with $\bar{x}$

2. Median Imputation

Median imputation replaces missing values with the median of the non-missing values in the same variable. This method is particularly useful when the data are skewed or contain outliers, as the median is a more robust measure of central tendency compared to the mean.

The median imputation value is computed as

$$\tilde{x} = \text{median}(X_{\text{obs}}) \dots\dots\dots (2)$$

where X_{obs} denotes the set of observed values. Each missing value is substituted with $\tilde{x}$

Multiple Imputation

Another method is multiple imputation is a statistical technique used to handle missing data. Instead of filling in a single value for each missing data point, multiple imputation fills in multiple plausible values to reflect the uncertainty about what the true value might be. This allows for more accurate and valid statistical inference.

Data Description

Functional data consists of information observed over a continuum, such as time, space, or frequency. Unlike traditional data types, it is represented by smooth functions or curves. This data is continuous, high-dimensional, and often exhibits strong internal correlation. It is typically observed with noise, requiring smoothing techniques for analysis. Functional data is commonly found in areas like environmental monitoring, biomedical signals, and economics. Its analysis involves special methods collectively known as Functional Data Analysis (FDA).

In the present research project, we utilized two inbuilt functional datasets, namely the Growth Data and the Phoneme Data, to illustrate and analyze key concepts in Functional Data Analysis (FDA). These datasets are widely used as benchmarks in the FDA literature due to their rich structure and diverse applications.

The Growth Data comprises longitudinal height measurements of boys and girls recorded over a range of ages. This dataset serves as a classic example of sparsely sampled, yet smooth functional data and is particularly well-suited for investigating growth patterns and gender-based differences in development trajectories.

The Phoneme Data, on the other hand, consists of log-periodograms derived from recordings of different phonemes (speech

sounds) spoken by male speakers. Each curve corresponds to a frequency spectrum, making this dataset a suitable example of densely observed, high-dimensional functional data. It is commonly used for classification and clustering tasks in FDA. The analysis of the Phoneme Data in this study was conducted using the `fda.usc` package, which provides a comprehensive suite of tools for functional data representation, transformation, and statistical inference.

These two datasets were selected due to their contrasting structural characteristics and analytical requirements, enabling a robust comparison of FDA methods in both sparse and dense observational contexts.

Proposed Method

In this analysis, we use Gaussian Process (GP) regression to impute missing values in functional data, which consists of smooth curves and covered both high and low dimensional data sets measured over time for multiple subjects. We simulate up to 40% missing values at random and use Gaussian Processes with a Radial Basis Function (RBF) kernel to predict and impute these missing values. Below is the methodology of our work.

Step 1: Understanding the data

We observe curves over time. Let $t_1, t_2, \dots, t_T$ be the curves over time. For each i subject, we have $Y_i(t_j)$ values.

Functional data means: Instead of isolated points, we think of each subject's measurements as a smooth function $Y_i(t)$

Step 2: Missing the observations

Some $Y_i(t_j)$ values are missing (we removed up to 40% randomly).

Goal: Fill in these missing values to reconstruct the full curve $Y_i(t)$

Step 3 : Using Gaussian Process (GP) Regression

We assume each curve is $Y_i(t)$ a random function drawn from a Gaussian Process, which means:

$$Y_i(t) \sim \mathcal{GP}(0, k(t, t')) \dots\dots\dots (3)$$

Here,

$k(t, t')$ is the kernel function that measures similarity between time points and .

We use the Radial Basis Function (RBF) kernel:

$$k(t, t') = \sigma^2 \exp \left(-\frac{(t - t')^2}{2\ell^2} \right) \dots\dots\dots (4)$$

where:

σ^2 -controls the overall variance,

ℓ -controls smoothness (how fast correlation decays as points get farther apart).

Step 4: The imputer

For each subject i :

Let $\mathbf{x}_{\text{obs}}$ be the observed time points (where y is known).

Let $\mathbf{y}_{\text{obs}}$ be the observed values at these points.

Let $\mathbf{x}_{\text{miss}}$ be the missing time points.

We use the observed data to predict missing values $\mathbf{y}_{\text{miss}}$ by:

$$\hat{\mathbf{y}}_{\text{miss}} = K(\mathbf{x}_{\text{miss}}, \mathbf{x}_{\text{obs}}) \cdot K(\mathbf{x}_{\text{obs}}, \mathbf{x}_{\text{obs}})^{-1} \cdot \mathbf{y}_{\text{obs}} \dots\dots\dots (5)$$

$K(\cdot, \cdot)$ are kernel matrices measuring similarity between points.

This gives the best guess (posterior mean) of missing values, using correlations learned from observed data.

Algorithm

Missing Value Imputation in Functional Data Using Gaussian Process:

This algorithm outlines the steps to impute missing values in functional data using Gaussian Process Regression (GPR). The process is applied to inbuilt datasets such as growth data and phoneme data, with simulated missing values at various levels: 1%, 2%, 5%, 10%, 20%, and 40%.

Step 1: Load Required Libraries

- Load the necessary R packages for functional data analysis and Gaussian process modelling.

Step 2: Load Functional Dataset

- Load the inbuilt dataset:

For growth data, use `growth$hgtf` or `growth$hgtn`.

For phoneme data, use `phoneme$learn$curves`.

Step 3: Introduce Missing Values

- For each selected percentage of missingness (1%, 2%, 5%, 10%, 20%, and 40%):
- Randomly select data points across the curves.
- Replace the selected values with NA to simulate missingness.

Step 4: Fit Gaussian Process Model

For each curve with missing values:

- Identify the observed (non-missing) time points.
- Fit a Gaussian Process Regression model using observed values.
- Predict the missing values using the trained Gaussian process model.
- Step 5: Reconstruct the Curves
- Combine the observed and imputed values to reconstruct the complete curves for each level of missingness.

Step 6: Evaluate Imputation Performance Using RMSE

- To measure the accuracy of the imputation:
- Compare the imputed values with the true (original) values at the same missing positions.
- Record RMSE for each missingness level to analyse the performance of the imputation method.

Root Mean Square Error

Root Mean Squared Error (RMSE) is a widely used metric to evaluate the accuracy of predictive models, especially in regression and imputation tasks. It measures the average squared difference between the observed true values and the values predicted or imputed by a model. RMSE is the square root of the Mean Squared Error (MSE) and is mathematically defined as:

$$\text{RMSE} = \sqrt{(1/n) \times \sum (y_i - \hat{y}_i)^2} \quad \dots\dots\dots (6)$$

Where:

y_i = actual (true) value

$\hat{y}_i$ = predicted or imputed value

n = number of observations

Purpose and Interpretation

RMSE gives a single number that represents the overall difference between imputed and true values.

It is expressed in the same unit as the original data, making it easy to interpret.

A lower RMSE indicates that the imputed values are closer to the true values, suggesting a better imputation method.

RMSE is sensitive to outliers due to squaring of errors, which helps in detecting large deviations in imputation.

In this report, RMSE is used to compare the performance of three different imputation methods applied to functional data with artificially introduced missing values at levels of 1%, 2%, 5%, 10%, 20%, and 40%. For each method, the imputed values were compared to the original (complete) data at the positions where missing values were introduced. RMSE was calculated and recorded for all levels of missingness, allowing a quantitative comparison of imputation accuracy among the three techniques.

Results

Why Graph?

In functional data analysis (FDA), graphs play a crucial role in visualizing the structure, patterns, and effectiveness of imputation techniques.

1. Visualize Original vs. Imputed Curves:

Graphs help us visually compare the original (complete) curves, the curves with missing parts, and the imputed versions. This makes it easier to understand how well the imputation method has worked

2. Understand the Pattern of Imputation:

Using graphs, we can see if the imputed parts look smooth and natural, and whether they follow the same trend as the observed data.

3. Support Accuracy Metrics:

Even though we use numerical values like RMSE to measure error, graphs give us a clear picture of where and how much error is present.

4. Check Kernel Behavior:

Since we use a Gaussian kernel, graphs help us observe whether the kernel smoothing works well across the whole curve or just in certain parts.

Graphs are essential in functional data imputation because they help us visually evaluate the success of the method, validate model assumptions, and communicate results effectively.

I. Graphical Comparison

(A) Graphs for growth data

1. 1% missing values

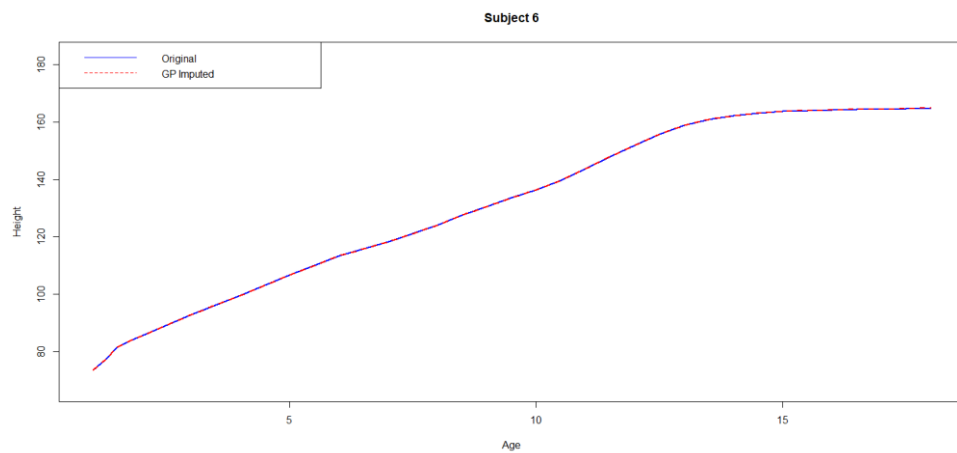


Fig.A:1

2. 10% missing values

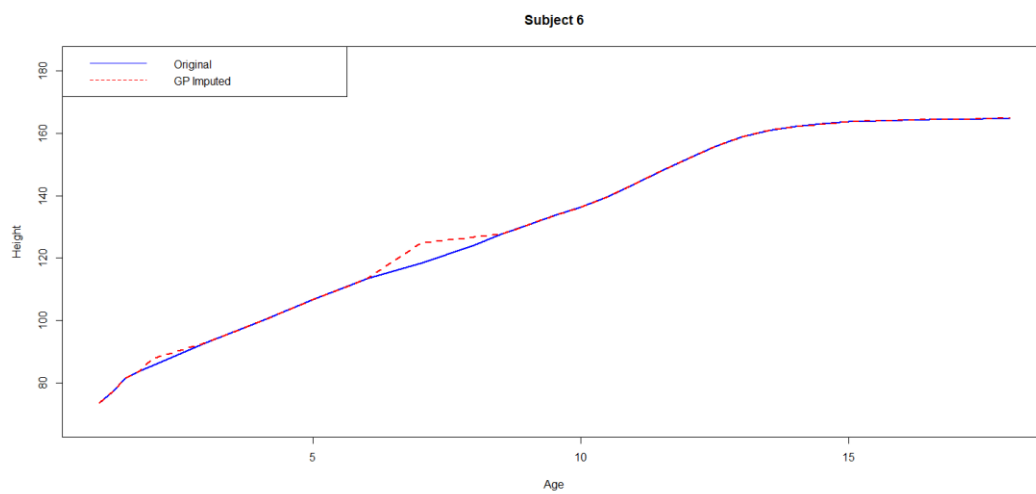


Fig.A:2

3) For 20% missing values

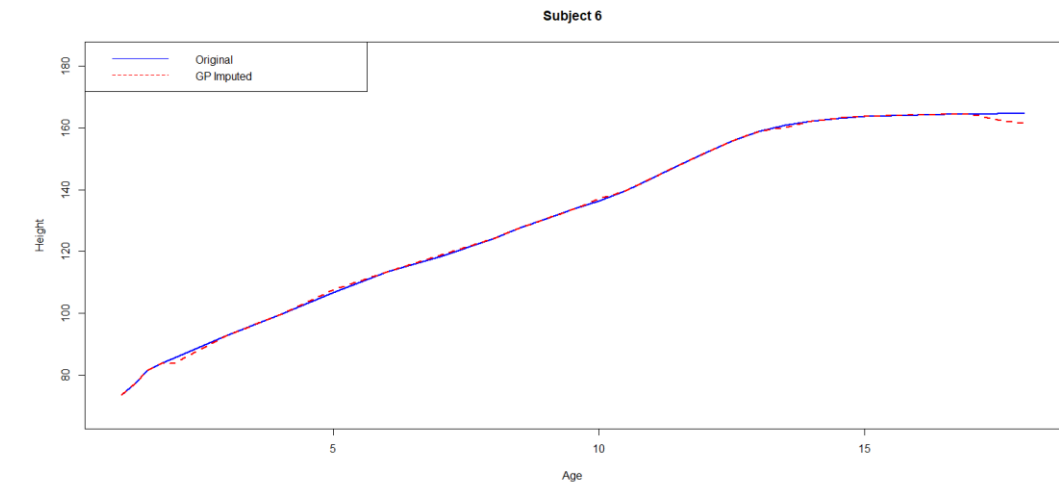


Fig.A:3

4) For 40% missing values

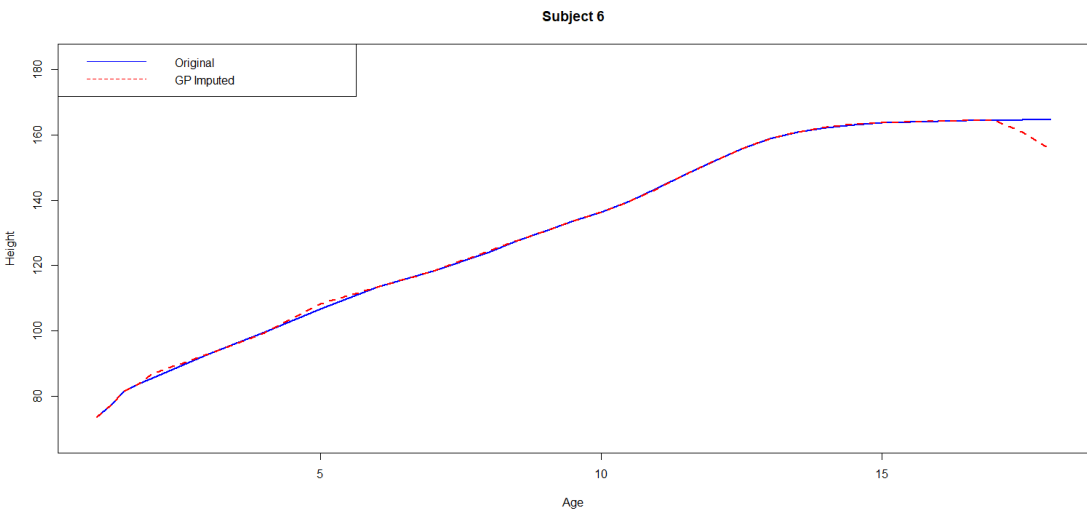


Fig.A:4

(B) Graphs for Phoneme data

1) 1% missing value

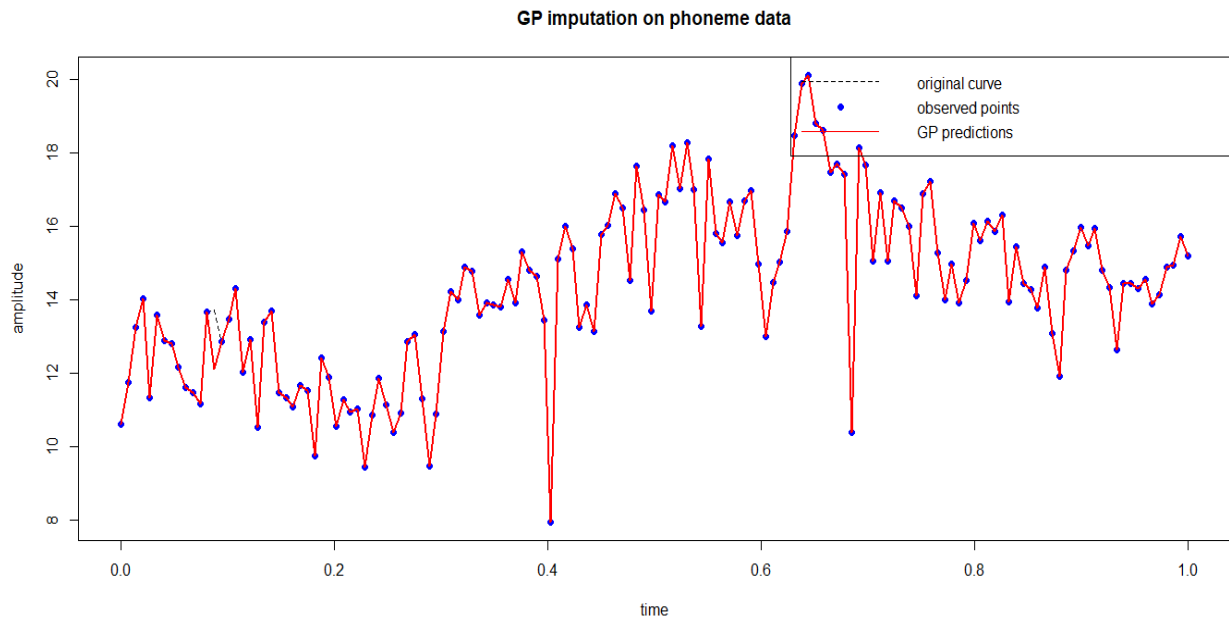


Fig. B:1

2) 10% missing values

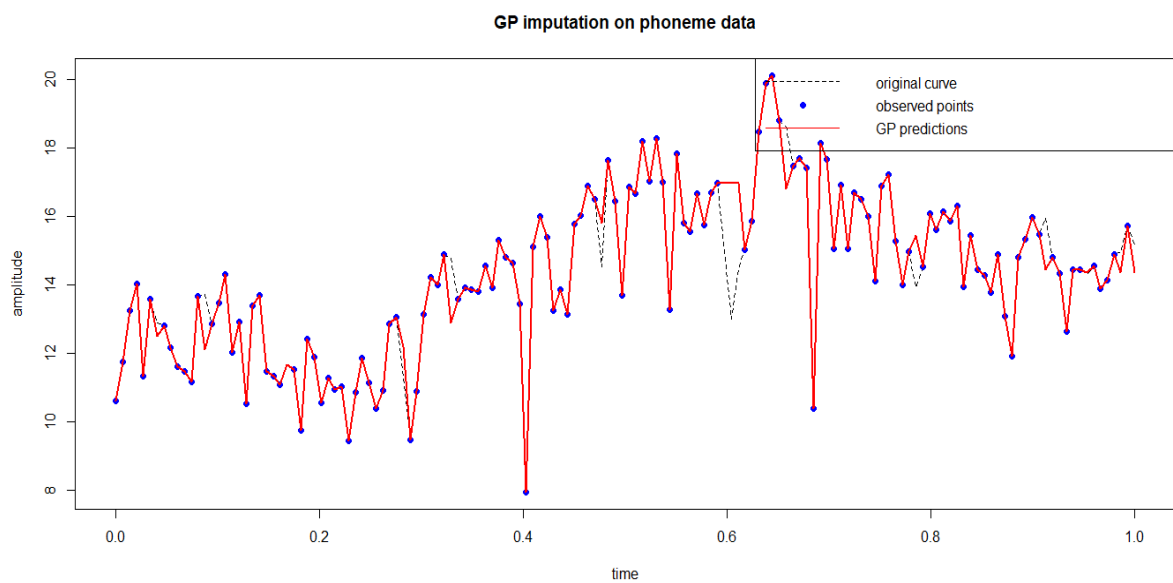


Fig. B: 2

3) 20% missing values

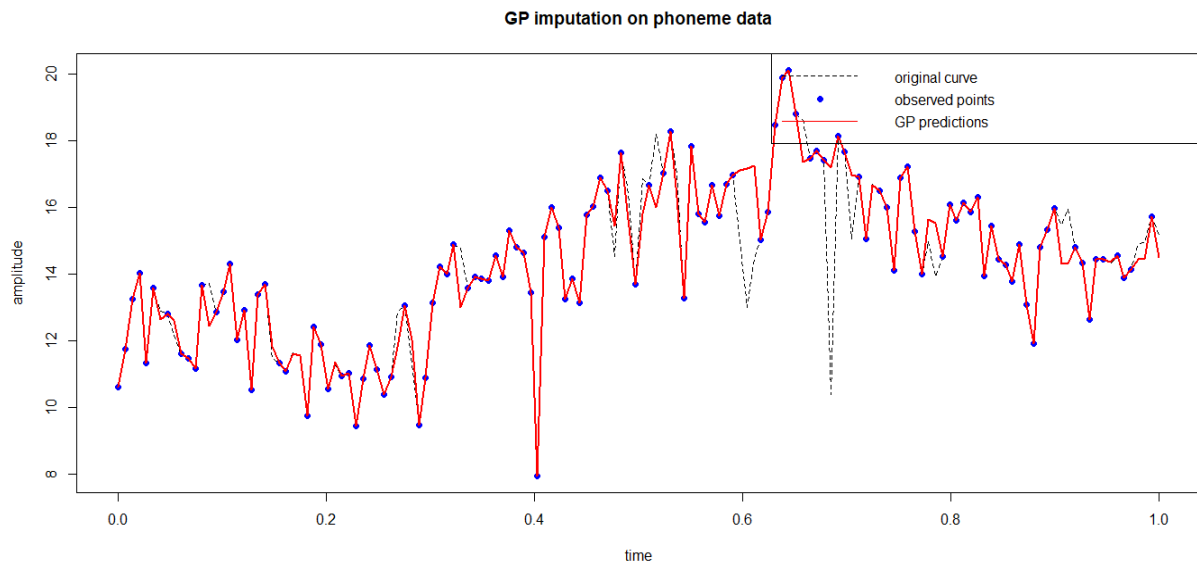


Fig.B:3

4) 40% missing values

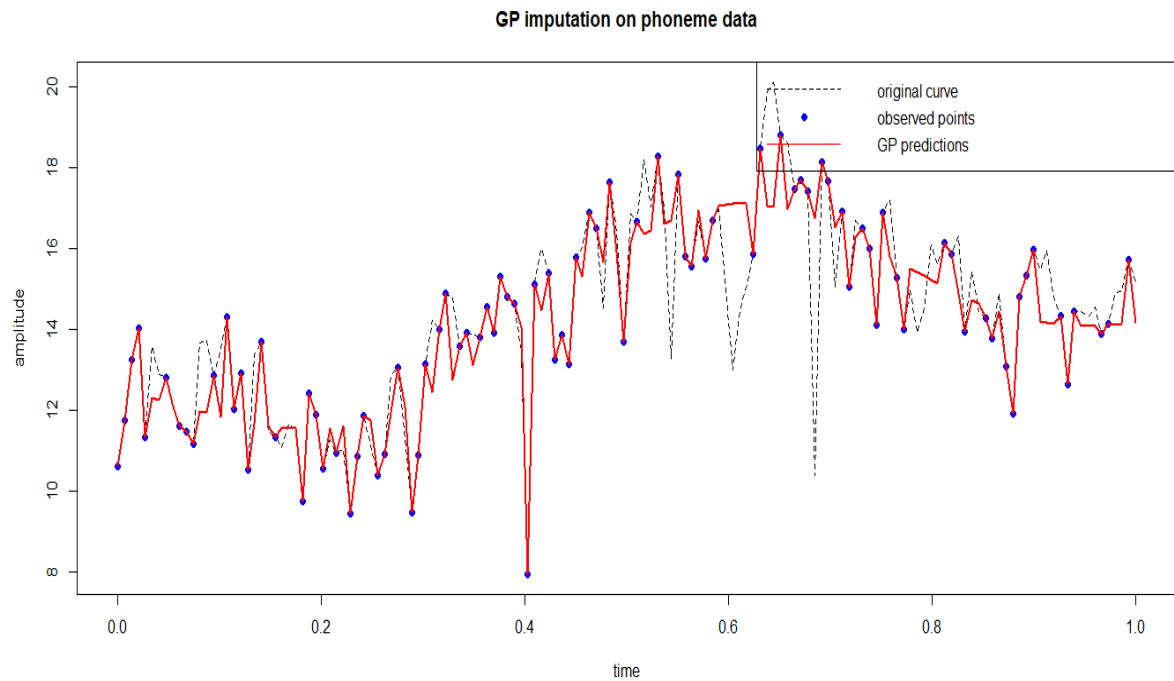


Fig.B:4

Interpretation :

From above graphs we see that for both the data sets ,1% missingness we have totally overlapped curves giving high accuracy furthermore for 10%we may find minimum and acceptable deviations ,for 20% more noticeable deviation from the original curve can be seen, especially in larger missing segments.GP show moderate accuracy and the gap between true and imputed curves becomes more significant for 40%. Some parts of the imputed curves may not match the actual trend. But still outperforms better than mean and Median imputations.

II. Comparison by tables

RMSE Tables

For Growth rate

Table 1:

Missing %	Mean Imp	Median Imp	GP Imp
1	25.8806	24.9428	0.5275989
10	31.5063	32.2936	2.65607
20	31.5789	32.1830	6.65607
40	31.8528	33.5307	6.552231

For Phoneme data

Table 2

Missing %	Mean Imp	Median Imp	GP Imp
1	0.5	10.56	1.58
10	1.69	6.95	0.242
20	2.09	7.08	0.259
40	2.19	7.35	0.182

Interpretation :

From the above tables for both data sets we conclude that the RSME value for GP Imp is better than Mean Imp and Median Imp.

Conclusion

Based on the graphs and tables discussed, it is clear that GP Imp outperforms the other imputation techniques evaluated. Although Mean Imp is a widely used approach for dealing with missing data, it consistently yields the highest RMSE across all levels of missingness. This result suggests that Mean Imp often leads to less accurate and less dependable data restoration, contributing to more significant errors in the dataset.

While Median Imp performs better than Mean Imp, it still does not reach the accuracy and dependability shown by GP Imp. Median Imp produces average results in various missing data scenarios, indicating that although it can manage a certain extent of missingness, it may not be the optimal choice when higher accuracy is essential.

GP Imp, on the other hand, shows superior performance by achieving the lowest RMSE at different missing data rates. This indicates that GP Imp is not only more accurate but also more reliable in maintaining data integrity. Its consistent advantage over other techniques under varying conditions demonstrates its robustness and makes it a strong candidate for situations where data quality is paramount.

In summary, the analysis highlights that GP Imp is the most dependable approach for imputing missing values, offering minimal error and high consistency. These qualities make it the preferred method for addressing missing data across diverse datasets. This comparison emphasizes the importance of selecting an effective imputation strategy to improve the accuracy and credibility of statistical analysis and forecasting.

Appendix

Packages required in R

ggplot2- For data Visualization

kernlab- For kernel method

fda - For functional data analysis

reshape2- For reshaping the data frames

DiceKriging- For modelling smooth curves and imputations